

Personal networks

The net – not that ridiculous Sandra Bullock movie, the actual Internet – depends solely on networking. This you know already. However, there's more to the net now than ever before, because everything from smart ACs to phones, tablets to PCs, cars to satellites all are hooked into the most complex digital organism ever created by man.

You may just want to do a Google search on your phone, but it's your mobile phone service provider's cell tower that transfers the request to a base station, which in turn sends it down a pipe to the IP network (queries DNS servers and the like), then to the closest Google server IP, queries the search term, and returns results back the way it came. If you're driving and switch cell towers, the network is aware of it and sends the data back to the right tower and to you. All of this happens in the blink of an eye.

However, this is the day and age of home networks and personal networks. Regardless of whether you're connected to the Internet, we're hooked over LAN or Wi-Fi to many other gadgets in our home/office. A printer, for example, or a shared folder, a NAS device, your smart TV, or DLNA compatible hardware – all of these are home networks.

What's more, we're already into the areas of personal area networks, and even body area networks (for wearable devices), and we're only going to have more and more complex yet local networks, which can always connect to larger ones. Perhaps this is why we find it best to refer to the evolution of the Internet (or all networked devices in general) as an organism, with a life force of its own.



You never know when you're being monitored

The Cloud

Without a doubt, the future seems to hold for us the most invisible storage possible. Cloud services depend solely on networking to offer you a space of your own – somewhere in the world, split across servers, just chunks of data in cyber space, storing all that you hold dear. Already so much is stored in your email account, or on Facebook and the likes and even more of your life is bound to be offloaded from your drives and memory cards on to the great unknown. Networking is what makes it all possible – even social networking is just another form of us going back to our natural roots, and bringing the concept of networking around full circle, to whence it began, billions of years ago.

Identification

One of the most important aspects of living in this technological marvel of a world is to know that you are indeed you. You wouldn't want someone else being able to withdraw all your money, or someone to use your name and face to commit a crime. Billions of dollars are spent trying to make fool-proof ways of ensuring that all the technologies you interface with know exactly who you are. Actually, it's not just people, but trillions of parts, goods, crates, devices, etc., needing to be identified correctly.

Take for example the case of when you fly to a new destination. You check in a bag perhaps, and since you're one of several thousand people flying out of your airport to one of several hundred destinations, the odds are staggeringly in favour of you ending up at your destination without your luggage. However, "Whoops! Sorry, but your bags got sent to Timbuktu. We'll have them sent to you in about 33 days", is

not something you're likely to hear, ever. Thanks to identification technologies, billions of people, goods, parts and thin-gamajigs all get to the right place purely because of technology. Let's look at some of the many fascinating ways in which this happens.

RFID

Short for Radio-Frequency IDentification, these are tags that can be attached to almost anything. From automobile spare parts to people, pets, foodstuff – if you can touch it, you can tag it. When your car with a tag drives through a sensor-equipped toll booth, your account is automatically debited. Another variation is EAS (Electronic Article Surveillance) systems, which basically prevent you from shoplifting.

RFID tags are of two types – passive or active. Passive tags need an active reader, and vice versa. The passive tag has no battery power, and it's the tag reader that activates the tag and reads it. Active tags are usually battery powered, and emit signals of their own, which even an active reader can pick up. Obviously active tags offer a greater range with regards to the distance at which they can be read, and this makes them ideal for use in larger transport systems. They're also commonly found in corporate offices as employee cards that are used to sign you in and out of buildings, or even allow access to some levels, and disallow access to where you're not authorised to enter.

Passive tags, obviously, are cheaper and smaller, but with a shorter working range and find their way into more common use across malls and retail outlets. Also smaller shipment items, such as those within courier companies, use this system to simplify tracking.

In the not so distant future, RFID will make life easier for all of us. Imagine filling up your shopping cart and just walking up to a payment area. Your bill will already be prepared, and all you need to do is pay. You only unload the cart at your car, and won't have to wait in long queues while a disinterested clerk takes ages scanning each item. There's even talk of RFID embedded in your body, much like some countries do with pets these days, to give you relevant access, or establish your presence without any visible action. Truly the

Handling baggage and training mice



Baggage handling is an arduous task that airports somehow manage to execute with aplomb. There is a lot of hard work

that goes into the design of automated baggage handling systems and the folks at Helsinki airport were kind enough to supply a video. Read about all this and mouse pheromones and more at <http://dgit.in/CSJuly2>